

# Harish Kumar A

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## Professional Summary

Robotics Software Engineer with 1+ year of hands-on experience developing and testing real-world AGV/AMR systems. Experienced in Python, C++, ROS 2, Linux, autonomous navigation, localization, path planning, motion control, computer vision, sensor integration, and embedded systems. Hands-on with Raspberry Pi, ESP32, LiDAR, IMU, encoders, cameras, motor controllers, and robot field testing, with strong hardware-software debugging and root-cause analysis skills.

## Technical Skills

**Programming:** C++, Python, C, Java

**Robotics:** ROS 2, AGV/AMR, Autonomous Navigation, Localization, SLAM, A\* Path Planning, Motion Planning, Motion Control, Sensor Fusion

**Kinematics & Control:** Differential-Drive Kinematics, Forward/Inverse Kinematics, Heading Control, Velocity Control, Closed-Loop Turning, Trajectory Analysis

**Perception:** OpenCV, YOLO, AprilTags, Depth Cameras, LiDAR, IMU, Encoders, Fiducial-Marker Localization

**Embedded:** Raspberry Pi, ESP32, Arduino, Teensy, UART, I2C, GPIO, Serial Communication, Motor Drivers, Stepper Motors, systemd, Power Sequencing

**Tools:** Linux, Git, GitHub, FreeCAD, Webots, SSH, VNC, PyQt6, Qt WebEngine, Qt WebChannel, JavaScript

## Experience

### Software Engineer

Mar. 2026 – Aug. 2026

*Lambda Robotics*

Kovilpatti, Tamil Nadu

- Develop and test Python/C++ software for autonomous AGV/AMR platforms, covering navigation, localization, motion control, and embedded communication.
- Integrate LiDAR, IMU, cameras, encoders, motor drivers, and embedded controllers across physical robotic systems and validate behaviour through field testing.
- Build Raspberry Pi-ESP32 communication for velocity commands, diagnostic telemetry, localization correction, and coordinated robot control.
- Diagnose sensor, communication, motor, heading, networking, and motion issues through systematic hardware-software debugging and root-cause analysis.

### Software Engineer Intern

Sep. 2025 – Feb. 2026

*Lambda Robotics*

Kovilpatti, Tamil Nadu

- Contributed to robotics prototypes involving embedded systems, sensors, motors, CAD, computer vision, and autonomous navigation; developed a Wi-Fi-controlled obstacle-avoiding robot.

## Selected Robotics Projects

### Vision and IMU-Based Autonomous Guided Vehicle

Mar. 2026 – Present

*AprilTags, IMU, A\* Path Planning, Differential Drive*

- Built and tested a differential-drive AGV for indoor waypoint navigation using AprilTag localization, IMU heading feedback, and Raspberry Pi-ESP32 communication.
- Implemented A\* path planning, heading correction, closed-loop turning, motion profiling, and diagnostic telemetry; achieved stable navigation at approximately 0.245 m/s.

### AGV Human-Machine Interface with Emotion-Based Interaction

Jul. 2026 – Present

*PyQt6, Qt WebEngine, Qt WebChannel*

- Designed a modular AGV HMI for task assignment, robot status, mission monitoring, battery, connectivity, and interaction using reusable PyQt6 components and bidirectional Python-JavaScript communication.

### Autonomous Mobile Service Robot for Cafe Applications

May 2026

*ROS 2, LiDAR, Depth Camera, OpenCV, YOLO, SLAM*

- Integrated LiDAR and depth-camera data with ROS 2 SLAM for indoor navigation and applied YOLO-based person/obstacle detection for speed reduction and stopping behaviour.

### Six-Axis Robotic Arm with Cycloidal Gear Reduction

Feb. 2026

*Kinematics, Mechanical Design, Robotic Manipulation*

- Contributed to a six-axis robotic-arm concept and evaluated compact cycloidal transmission designs; studied forward/inverse kinematics, workspace, singularities, joint motion, and load conditions.

## Education

### B.Tech in Artificial Intelligence and Data Science

2022 – 2026

Dr. Mahalingam College of Engineering and Technology, Pollachi, Tamil Nadu

CGPA: 8.14/10